

1. a) The progressive wave with an amplitude of 4 cm, period of 0.1 s and wavelength of 0.12 m is moving to the right.

- i. Calculate
 - a) the frequency
 - b) the angular frequency
 - c) the wave number

- ii. Write the wave equation.

[4 marks]

- b) Two identical waves are travelling towards each other ;

$$y_1 = 8 \sin \left(\frac{3}{5} \pi t - \frac{1}{3} \pi x \right)$$

$$y_2 = 8 \sin \left(\frac{3}{5} \pi t + \frac{1}{3} \pi x \right)$$

- i Name the type of produced wave.
- ii Write the equation of the produced wave .
- iii Calculate the amplitude of the particle at $x = 2.5 \text{ m}$.
- iv Calculate the wavelength of the wave.
- v. Calculate the speed of the wave.

[12 marks]

- 2 a) A copper wire has a radius of 0.2 mm and a length of 8 m. Given the resistivity of copper is $1.72 \times 10^{-8} \Omega \text{ m}$.

- i. Calculate the resistance of the copper wire.
- ii. If a charge of 120 C passes through the wire within 1 minute, calculate the current that flows through the wire during that time.

[4 marks]

- b) Four resistors are connected to a battery as shown in **Figure 1**. The resistance of R_1 and R_2 is 2Ω while R_3 and R_4 is 4Ω . The voltage supply is 12 V.

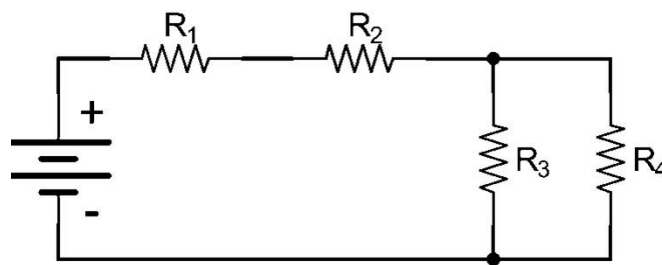


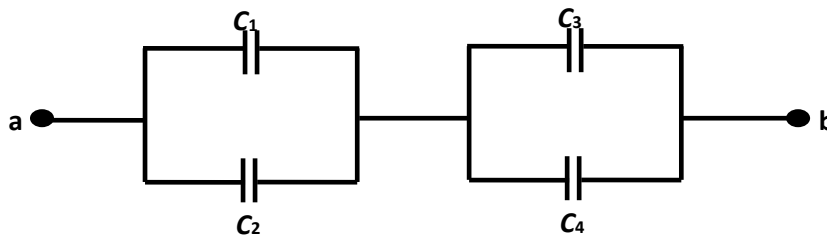
Figure 1

Calculate the

- i. effective resistance of the circuit
- ii. total current in the circuit
- iii. voltage across resistor R_1

[8 marks]

3.

**Figure 2**

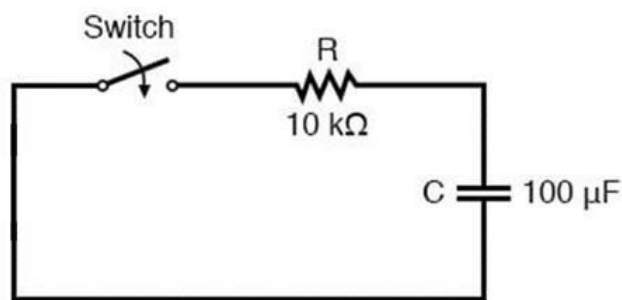
- a) **Figure 2** shows four identical capacitors of $5\ \mu\text{F}$ connected between points a and b. Total charge in the circuit is $6 \times 10^{-5}\ \text{C}$.

Calculate the

- i. effective capacitance.
- ii. voltage supply across points a and b.

[4 marks]

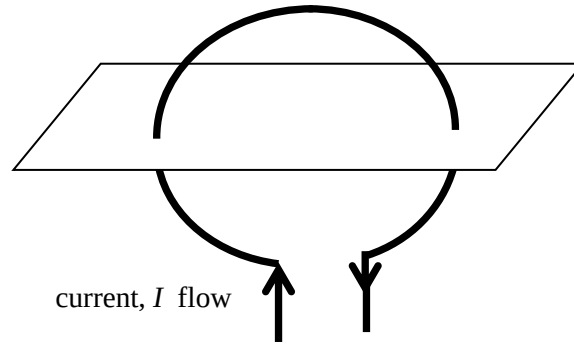
- b) **Figure 3** shows a circuit of capacitor and resistor. The capacitor is fully charged. When the switch is closed, the capacitor is discharging.

**Figure 3**

- i. Calculate the
 - a) time constant
 - b) time taken for the charge reduced to 25% of its maximum value
- ii. Sketch a graph of charge on the capacitor against time.

[6 marks]

4. a)

**FIGURE 4**

A student is provided with 2.0 m long wire. Then, the wire is bent into a circular coil of one turn as shown in **FIGURE 4**. A current of 0.15 A is flowing through it.

- (i) Sketch the magnetic field lines produced by circular loop.
- (ii) Calculate the radius of the circular coil.
- (iii) Calculate the strength of the magnetic field at the centre of the coil.

[7 marks]

b)

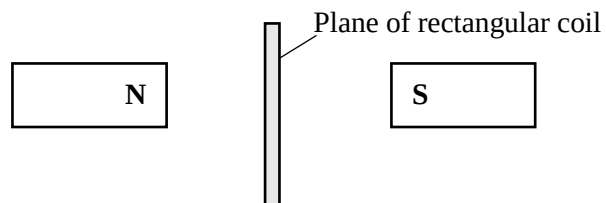
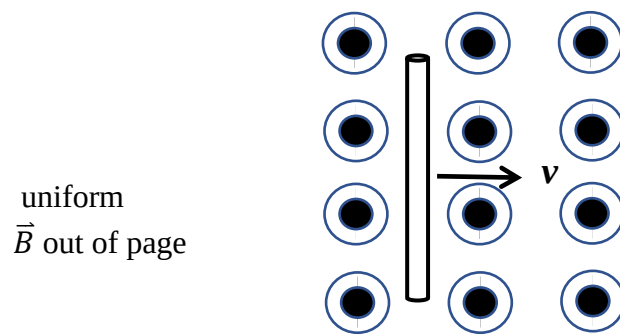
**FIGURE 5**

FIGURE 5 shows the plan view of a rectangular coil in a magnetic field. The area of the rectangular coil is 5 cm² and the magnetic field strength is 1.5 T. Calculate the magnetic flux through the coil.

[2 marks]

c)

**FIGURE 6**

A 30 cm long metal rod AB is moved at speed of 50 m s^{-1} across a uniform magnetic field strength of 0.25 T. The motion of the rod is perpendicular to the magnetic field as shown in **FIGURE 6**.

- (i) Calculate the motional induced emf in the rod.
- (ii) If the rod is connected in series to the resistor of resistance 10Ω , calculate the induced current.
- (iii) State the direction of the induced current.

[5 marks]

5. a) A pencil of height 12 cm is placed at the centre of curvature of a concave mirror with focal length 30 cm.
 - (i) Calculate the position of the image.
 - (ii) Calculate the image size.
 - (iii) State two (2) characteristics of the image.
 - (iv) Sketch and label a ray diagram to show the formation of the image.

[8 marks]

b) An inverted image is produced at a distance of 40 cm from an object by a convex lens. If the magnification is 4, calculate the

- (i) object distance.
- (ii) image distance.
- (iii) focal length of the lens.

[6 marks]

6. a)

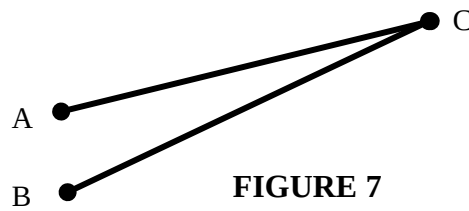
**FIGURE 7**

FIGURE 7 shows two paths of coherent lights from points A and B that produce an interference pattern at point C. Determine whether it is a constructive or destructive interference if AC and BC are 2.2λ and 5.7λ respectively.

[2 marks]

b)

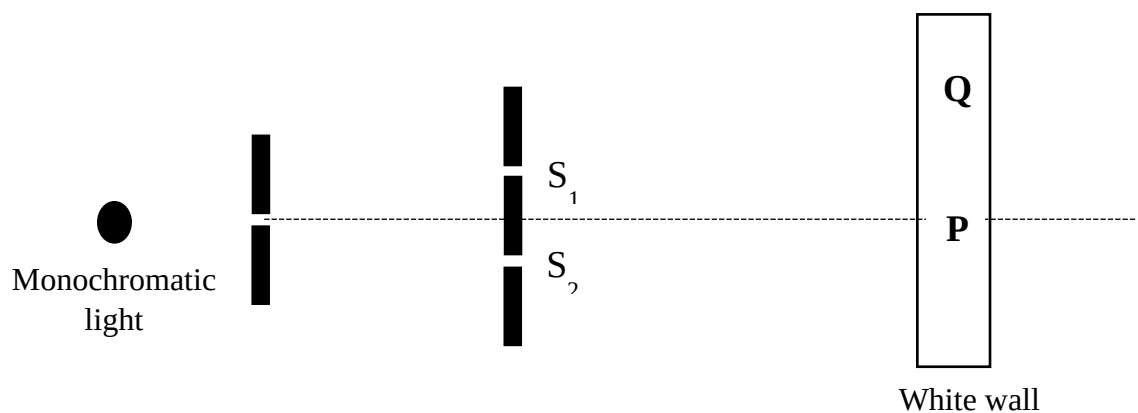
**FIGURE 8**

FIGURE 8 shows a monochromatic light passes through two narrow slits, S_1 and S_2 that are 0.1 mm apart. The interference pattern is observed on a white wall 1.2 m from the slits. A bright fringe is located at point P and a dark fringe is observed at point Q. The separation between two adjacent bright interference fringes is 6.0 mm.

- (i) Calculate the wavelength of the light.
- (ii) Calculate the distance of the fourth bright fringe from point P.
- (iii) Calculate the separation between the zeroth-order minimum and first order maximum.
- (iv) When a light source of wavelength 560 nm is used, the 40th dark fringe is observed at Q. However when a light source of wavelength, λ is used, the 35th dark fringe is observed at Q. Calculate the value of λ .
- (v) What is the change that can be observed if the distance between coherent sources is decreased?

[12 marks]

